Syngamy

Report:	c841f58e
Generated:	2026-07-31 03:29:19Z
Run started:	not recorded
Software:	OriginMarker 2.1.1 (Kinetochore), Syngamy module
Sperm donor:	GSM4472397_sperm_DNA_71.CEL.txt.gz (825,656 markers, SHA-256 1b56911dae7030f4)
Oocyte donor:	none supplied, so a maternal origin is inferred from paternal absence rather than measured
Donor heterozygosity:	16.658% autosomal, of called markers
Samples:	1 analysed (1 unclear)
Assembly:	GRCh37
Data handling:	every file was read in the browser that produced this report. No genotype was transmitted, and none is retained.

Result summary

Sample	Origin	Absence	Ceiling	x ceiling	Zygosity	Sperm	Informative
GSM4774687_pMII-8_61.CEL.txt.gz	unclear	3.51%	7.88%	0.4x	unknown	Y-bearing	383,652

Absence is the rate at which the donor's obligate allele is missing where he is homozygous. Ceiling is what this sample's own noise can manufacture: its no-call rate times its heterozygous fraction. A sample is called as lacking a paternal contribution only above 3x that ceiling, and as carrying one only at or below it; in between it is left uncalled. No confidence percentage is reported, deliberately: see Interpretation.

GSM4774687_pMII-8_61.CEL.txt.gz vs GSM4472397_sperm_DNA_71.CEL.txt.gz

UNCLEAR

Not separated by the available evidence.

absent 3.51% | ceiling 7.88% | 0.4x | 383,652 informative markers | zygosity unknown | Y-bearing sperm

Evidence

Measurement	Observed	Reference	The reference is
Donor alleles absent	3.51%	7.88%	this sample's no-call rate times its heterozygous fraction. Dropout manufactures absence only by turning a heterozygous call homozygous and discarding the donor's allele, so the bound is the product and neither term alone.
Alleles the donor lacks	23.93%	8.33%	half the donor's heterozygosity. A second parent supplies an allele he lacks only where he is homozygous, at sum(pq), and his heterozygosity is sum(2pq).
Heterozygous BAF band	18.06%	8.00%	measured at 15-16% for diploid genomes and 1.3-3.4% for uniparental ones. Dropout does not mimic this: a biparental embryo at 33% dropout still reads 22.2%.

Sex chromosomes

Chr	Informative	Absent	Rate	x ceiling	Verdict	Note
chrX	8,757	3,277	37.42%	4.7	expected absent	no paternal X: this sample carries the parent's Y
chrY	719	24	3.34%	0.4	present	

Chromosomes

Chr	Informative	Absent	Rate	x ceiling	Verdict	Note
chr1	30,469	1,610	5.28%	0.7	present	
chr2	31,327	615	1.96%	0.2	present	
chr3	23,791	1,173	4.93%	0.6	present	
chr4	23,940	472	1.97%	0.3	present	
chr5	22,814	458	2.01%	0.3	present	
chr6	27,223	637	2.34%	0.3	present	
chr7	21,298	558	2.62%	0.3	present	
chr8	17,169	797	4.64%	0.6	present	
chr9	16,476	722	4.38%	0.6	present	
chrX	8,757	3,277	37.42%	4.7	expected absent	no paternal X: this sample carries the parent's Y
chrY	719	24	3.34%	0.4	present	
chr10	18,142	585	3.22%	0.4	present	
chr11	20,537	822	4.00%	0.5	present	
chr12	18,640	535	2.87%	0.4	present	
chr13	11,088	719	6.48%	0.8	present	
chr14	13,622	352	2.58%	0.3	present	
chr15	12,907	202	1.57%	0.2	present	
chr16	15,345	332	2.16%	0.3	present	
chr17	13,644	844	6.19%	0.8	present	
chr18	11,479	197	1.72%	0.2	present	
chr19	13,016	1,035	7.95%	1.0	unclear	
chr20	8,895	389	4.37%	0.6	present	
chr21	6,124	46	0.75%	0.1	present	
chr22	5,706	368	6.45%	0.8	present	

Sample quality

Markers: 825,656 read, 488,055 called (59.1%)

Research use only. Parent of origin here is a statistical call from array genotypes and requires confirmation by an independent method in a qualified genetics laboratory. Not a clinical diagnostic.

No-call:	40.89%
Heterozygous:	24.96% of called
chrX / autosomal het:	0.114
Sex call:	male
Product:	UK Biobank Axiom Array
Assembly:	GRCh37 (0 placements illegal under GRCh37, 6,315 under GRCh38, 825,656 tested)
Allele coding:	ok BAF corroborates 0=AA, 2=BB
Mean BAF AA/AB/BB:	0.077 / 0.525 / 0.936
SHA-256:	36588244213aa3b600bb4a87275c2edb1139e6f79b2ea86644b69358a8625cef
File size:	63,325,418 bytes

Quality gates

Gate	Value	Verdict	Basis
call rate	59.1%	exclude	below 60%: excluded (Natesan 2014, the only published threshold measured on amplified material)
het-to-hom asymmetry valid	59.1%	exclude	SUSPENDED: below 60% call rate erroneous heterozygous calls are common, so a het call is no longer robust and the key/non-key partition loses its guarantee
genome-wide LOH	25.0%	report only	no genome-wide LOH signature
heterozygosity plausible	25.0%	usable	within the plausible range for one diploid genome
numeric genotype coding	-	usable	BAF corroborates 0=AA, 2=BB
sex call	11.4%	usable	chrX/autosome heterozygosity ratio 0.114 is inside the male band. Dropout cancels in the ratio, so this survives amplified material.

Limits on this call

Call rate 59.1% is below 60%, where erroneous heterozygous calls become common. Zygosity is read from the heterozygous band, so it is not reported here and the genome cannot be separated into uniparental or biparental. The presence or absence of this parent's contribution is unaffected: that is Mendelian and does not rest on heterozygous calls.

Research use only. Parent of origin here is a statistical call from array genotypes and requires confirmation by an independent method in a qualified genetics laboratory. Not a clinical diagnostic.

Methods

Parent of origin was determined from SNP array genotypes using OriginMarker 2.1.1 (Kinetochore). For each of 1 sample(s), the rate at which the sperm donor's obligate allele was absent was computed across autosomal markers where he is homozygous and the sample is called, and compared against the rate that sample's own genotyping noise can produce, taken as the product of its no-call rate and its heterozygous fraction. A sample was called as lacking a paternal contribution only where the observed rate exceeded that bound by 3-fold, and as carrying one only where it fell at or below the bound; rates between the two were left uncalled. Whether a second parent contributed was assessed from the rate at which the sample carries alleles the sperm donor does not possess, against half his heterozygosity, which is the contribution a second parent makes. Zygosity was taken from the fraction of B-allele frequencies falling in the heterozygous band, 0.35 to 0.65. Sperm type was inferred from the chrX rate rather than from chrY, so that an X-bearing sperm carrying a full paternal genome is distinguishable from an absent paternal contribution; neither case leaves a Y.

No oocyte donor array was supplied. A maternal origin below is therefore inferred from the absence of the paternal contribution rather than measured, and a sample belonging to neither declared parent is indistinguishable from one that is maternal in origin.

Assembly was determined from marker positions against the UCSC chromInfo and gap tables and found to be GRCh37. No liftOver was performed.

The following limits applied to this run and are reported rather than resolved: (1) Call rate 59.1% is below 60%, where erroneous heterozygous calls become common. Zygosity is read from the heterozygous band, so it is not reported here and the genome cannot be separated into uniparental or biparental. The presence or absence of this parent's contribution is unaffected: that is Mendelian and does not rest on heterozygous calls.

Interpretation

No confidence percentage is attached to a call, and this is deliberate. The per-sample likelihood ratio between "paternal genome present" and "absent" runs past 10^10000 on a genome-wide array, which is not a probability any reader should be handed. The honest figure is the empirical accuracy of the method, and on the pairs of known relationship it has been run against that is 9 of 9 correct, whose 95% lower bound is roughly 72%. About 300 consecutive correct calls would be needed to claim 99%. What is reported instead is the margin: the observed rate, the reference it was measured against, and the ratio between them, so a reader can see how far from the boundary the call sits.

The absence axis separates by roughly thirty-fold and carries the call. The alleles-the-donor-lacks axis separates by about 1.6-fold and only distinguishes a biparental genome from a paternal-only one that is heterozygous; treat a near-boundary call on that axis as provisional. A sample scoring in the uncalled band is not a weak positive, it is an absence of evidence: an array from the second parent measures dropout directly and settles it.

Constants and where they come from

Constant	Value	Provenance
Mendelian inconsistency floor	0.0063	Kothiyal 2019, 0.63% of variant sites across 1,314 nuclear families. A lower bound on the spurious-violation rate, never the value itself.
Paternal-absence noise bound	no-call rate x heterozygous fraction	Derived per sample, not fitted. Bounds every related pair measured, within 2x: 0.13% predicted against 0.05% observed, 0.18% against 0.16%, 10.4% against 9.69%.
Absence calling margin	3x	How far above the noise bound an absence must sit before it is called.
Residual absence floor	0.005	Absence on clean data from genotyping error alone, measured at 0.03% and 0.05%. Added to the bound so a near-perfect array still has a non-zero reference.
Second-parent signal	half the donor's heterozygosity	Derived. A second parent supplies an allele the donor lacks only where he is homozygous, at sum(pq); his heterozygosity is sum(2pq) over the same markers.
Diploid heterozygous BAF band	0.08 of markers in 0.35-0.65	Measured at 15-16% for diploid genomes and 1.3-3.4% for uniparental ones. Dropout does not mimic this: a biparental embryo at 33% dropout still reads 22.2%.
Per-chromosome minimum	200 informative markers	Below this a chromosome is not reported: the rate is too noisy to place against the ceiling.
Assembly tables	UCSC chromInfo and gap, hg19 and hg38	Freely redistributable. No liftOver chain file is used or shipped: those carry a non-commercial field-of-use restriction Apache 2.0 cannot sublicense.

Files read

File	Role	Bytes	Markers	SHA-256
GSM4472397_sperm_DNA_71.CEL.txt.gz	donor	63,674,062	825,656	1b56911dae7030f4ecebc2ade9c7b1bacee23def81d130e6b3d0ac9edb388169
GSM4774687_pMII-8_61.CEL.txt.gz	sample	63,325,418	825,656	36588244213aa3b600bb4a87275c2edb1139e6f79b2ea86644b69358a8625cef

The hash is of the file as read, so a reviewer can confirm the input without being sent it.

Citation

OriginMarker 2.1.1 (Kinetochore). Syngamy: parent of origin from SNP array genotypes. Report c841f58e, generated 2026-07-31 03:29:19Z.
Kothiyal P et al. Mendelian inconsistent signatures from 1,314 ancestrally diverse family trios. J Comput Biol 2019;26(5):405-419. doi:10.1089/cmb.2018.0253. Cited for the inconsistency floor above.